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Research Report

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1 Introduction

Annual reports filed to the U.S. Securities and Exchange Commission (SEC) under Form 10-K are widely considered to be among the most informative disclosures of public listed firms. Although traditional financial analysis focuses primarily on quantitative and accounting metrics such as earnings, leverage, and cash flows, these filings have been proven to contain extensive narrative sections that provide meaningful information about management's perspective on strategy, applied or potential risks, forward guidance, and performance. In particular, Item 1 (Business), Item 1A (Risk Factors), and Item 7 (Management's Discussion and Analysis of Financial Condition and Results of Operations — MD&A) provide structured textual disclosures across firms, making them comparable and suitable for large-scale empirical analysis.

Our research is inspired by the "Lazy Prices" (Cohen, Malloy & Nguyen, 2020) and (Hoberg & Phillips, 2016) papers. Following their methodological frameworks, we constructed a large-scale panel dataset linking 10-K textual features to firm-level financial and market data for S&P 500 firms over the period 2004–2025. Firm identities are linked across databases using WRDS identifier mappings (CIK, GVKEY, and PERMNO), and filings are retrieved from EDGAR and analyzed into sections. In order to ensure valid inference, the sample is restricted to yearly observations in which the firm is included in the S&P 500 and has an available 10-K filing, trying to tackle survivorship and look-ahead bias arising from previous disclosures, and ensuring that only information available at the time is used.

The empirical framework focuses on measuring strategic shifts thought changes in the semantic content of Section 1A and Item 7 ((Management's Discussion and Analysis of Financial Conditions) of 10K fillings. These Section are strategically chosen because they provide executives and managers the greatest flexibility to communicate future strategic choices and firms' risk, and thus reveal substantive shifts in strategic firm direction. While previous studies that focused on the industry strategic position, relied on dictionary-based methods like Bag-of-Words or TF-IDF cosine similarity which often fail to capture the semantic context of the documents, this study employs a dense document embedding generated by the all-MiniLM-L6-v2 transformer model. For each firm-year, a composite document embedding is computed by taking the mean of the embeddings generated for both Item 1A and Item 7. The Semantic Drift Scores which capturesthe magnitude of strategic repositioning is calculated as the complement of the cosine similarity between the composite embeddings of two consecutive years producing a value ranging from zero to one, with higher values indicating more intense strategic shift.

The resulting firm yearly panel is merged with stock returns and accounting variables from CRSP and Compustat, aligning each filing with subsequent returns to ensure a design

which looks forward. The predictive ability of the strategic shift textual signal is evaluated using two complementary approaches. Initially, portfolio sorts examine whether sorting firms into quintiles based on each signal produces differences in future returns. Second, Fama-MacBeth regressions across sections test whether the signals maintain their explanatory power after controlling for standard firm characteristics such as size, profitability, and momentum.

Our study contributes to the literature in several ways. First, we provide evidence on whether strategic shifts, captured through changes in the semantic content of Items 1A and 7 of 10-K filings, are reflected in subsequent stock returns. Methodologically, our study contributes to the literature by proposing an innovative alternative way of measuring strategic change at the firm level: applying a neural network-based sentence transformer model (`all-MiniLM-L6-v2`) to Items 1A and 7 of annual 10-K filings. Traditional industry classifications, such as SIC codes provided by Compustat, rarely change even when a firm's underlying business has substantially evolved, which makes them poorly suited to capture strategic repositioning. In contrast, the Semantic Drift Score proposed in this study is capable of detecting strategic shifts directly from the language managers use to describe their business and risks, rather than from static industry labels.

Similarly to (Cohen et al., 2020), our study is guided by one primary hypothesis that firms. Firms that exhibit higher strategic shift, as revealed by larger semantic shifts in Items 1A and 7 of their 10-K filings earn higher future 12 month returns than firms whose disclosures remain largely unchanged.

Combining the above, this framework provides an approach in evaluating the informational content of corporate disclosures. Combining a neural network-based measure of semantic drift with a carefully constructed, bias-controlled sample, the analysis aims to assess whether and how strategic shifts in narrative reporting are incorporated into asset prices.

2 Background and Related Literature

The application of Natural Language Processing (NLP) to 10-K corporate filings has expanded rapidly in recent years as computational power and the availability of textual data have increased

2.1 Textual data in financial economics

The use of textual data in financial economics stems from the idea that corporate disclosures contain valuable qualitative information that is not captured by accounting variables alone. Early evidence supports this view. F. Li (2008) shows that less readable annual reports are associated with lower future earnings, consistent with the notion that managers may obscure poor performance through complex language. Similarly, Tetlock (2007) and Tetlock, Saar-Tsechansky and Macskassy (2008) demonstrate that negative language in financial text can predict both earnings and stock returns, establishing that textual signals are informative, yet not fully incorporated into prices.

2.2 Bag-of-Words Approach (BoW)

The most widely adopted NLP technique in financial text analysis is the simple bag of words (BoW), which is commonly used for assessing sentiment. The BoW approach uses algorithms to simply categorize words based on different criteria by measuring the frequency of words or phrases, without considering the syntactic structure or the context of the sentences in which the words appear.

Early studies in the field, relied on general psychological dictionaries to analyze the tone of the corporate disclosures. The most popular dictionary for sentiment analysis back then was the Harvard IV-4 dictionary. However, several papers (Henry & Leone, 2016; F. Li, 2010), suggested later that dictionaries that are built for general purposes might not be suitable for sentiment analysis of corporate disclosures, without undermining the results of previous studies. A key methodological development in this literature is the introduction of specific domain sentiment measures. Loughran and McDonald (2011) show that general purpose dictionaries tend to misclassify financial language in a systematic base and propose a dictionary specifically focused in finance, which has since become the standard for measuring tone in corporate disclosures. Loughran and McDonald (2011) developed a static, domain specific dictionary which categorized words into negative, positive, uncertainty, litigious, strong modal, and weak modal and since then became the dominant dictionary in the field. Additionally, they demonstrated that approximately 75% of Harvard dictionary's negative words like tax or liabilities are misclassified words, that cannot be classified as negative in financial disclosures. This work figurates a sentiment

analysis in 10-K filings based on a dictionary and highlights the importance of tailoring linguistic tools to the institutional context needs.

Following the study of Loughran and McDonald (2011), the literature has expanded to thematic dictionaries. More specifically, Baier, Berninger and Kiesel (2020) created a ESG (Environmental, Social, Governance) dictionary while K. Li, Mai, Shen and Yan (2021) constructed a dictionary to measure corporate culture. The K. Li et al. (2021) dictionary is innovative as they use a semi-supervised machine learning approach and word embedding model (word2vec) to create the "culture dictionary".

2.3 Management Discussion and Analysis (MD&A)

The majority of the NLP literature focuses on the Management Discussion and Analysis (MD&A) section of 10-K filings as this is the only section where the managers have the flexibility to disclose information strategically, making it a valuable source of their intentions and future plans of the companies (F. Li, 2010).

Brown and Tucker (2011), focusing on yearly over yearly modifications in MD&A sections, find that intensity of changes in MD&A section are commensurate with the economic changes experienced by a firm. Additionally, they find that the MD&A section and 10-K filings in general have become less informative over the last years as managers rely more on boilerplate disclosures. Similar were the findings of K. Li et al. (2021), regarding 10K usefulness, who demonstrated that due to increased AI and Machine Learning readership in recent years, firms have shaped their filings in order to minimize negative sentiment for the stock. On the other hand, Cohen et al. (2020) demonstrated that major changes in 10-K still hold information value, by constructing a portfolio with long position on firms with unchanged fillings and short on firms with major changes in 10-K fillings, achieving a monthly alpha of 188 basis points.

2.4 Document Similarity Measures

Another approach that is widely adopted in NLP is to examine the similarity between documents. The most popular similarity measure is cosine similarity which calculates the cosine of the angle between the vector representations of two documents (Senave, Jans & Srivastava, 2023), producing values between 0 and 1. Identical documents will return a value of 1 while completely dissimilar documents will return 0. In terms of finance, Hanley and Hoberg (2012) used cosine similarity to measure the similarity between IPO prospectuses in order to examine firms' disclosure strategies and exposure to litigation risk. Hoberg and Lewis (2017) examine whether fraudulent firms produce abnormal disclosures by computing cosine similarity between a firm's MD&A disclosures and the average disclosure of its non-fraudulent industry peers with similar size and age.

Hoberg and Phillips (2016) construct text-based industry classifications from 10-K product descriptions and show that these outperform traditional SIC codes in capturing competitive relationships. Their findings suggest that shifts in disclosure language reflect changes in firm strategy and economic positioning, motivating the use of thematic text measures to track strategic dynamics over time.

Another important part of research focuses not on the level of disclosure, but on how it changes over time. Cohen et al. (2020) show that firms with highly similar consecutive 10-K filings tend to underperform, contrary to those who substantially revise their disclosures. Textual change, typically measured using TF-IDF cosine similarity, captures the extent to which new information is being introduced. This result is often interpreted through a limited-attention framework (Hirshleifer & Teoh, 2003), in which investors are slower to process complex qualitative information. As a result, measures of text change provide a simple but powerful proxy to disclosure information and form a central component of this project.

2.5 Advanced NLP Models: BERT and Topic Modeling

More recently, advances in NLP have introduced models based on transformers, which capture meaning dependent on context in a financial text. A fundamental innovation in the field of textual analysis is the BERT model, an unsupervised pre-trained model introduced by Devlin, Chang, Lee and Toutanova (2019). Unlike traditional BoW methods that treat words as independents units, ignoring the word order and context, Bert model generates contextual token representations able to capture the semantic meaning of whole sentences and paragraphs. Bert family's model, specifically the FinBert financial domain specific model, developed by Araci (2019) and Huang, Wang and Yang (2023), is widely used for sentiment classification and ESG topics, improving at the same time the approaches based on a dictionary, accounting for syntax, negation, and semantic context. Huang et al. (2023), analyzing the sentiment of 10-K, found that FinBert model achieves significantly higher accuracy over Loughran and McDonald dictionary as well as NB, SVM, RF, CNN, and LSTM algorithms. Also, embedding methods allow for expansion driven by data of keyword dictionaries and more flexible representations of textual similarity. But, while these methods increase predictive power, they also introduce a trade-off between interpretability and model complexity.

Another neural network model that belongs to the Bert family is the BertTopic model, a neural topic model developed by Grootendorst (2022). While traditional topic models such as Latent Dirichlet Allocation (Blei, Ng & Jordan, 2003) and Non-Negative Matrix Factorization (NMF) (Févotte & Idier, 2011) might fail to accurately capture the topic of a document as they ignore the context of words in a sentence, BertTopic Model overcomes these limitations by combining pre-trained transformer-based document embeddings with

a class-based TF-IDF. Jehnen, Teuteberg and Lucka (2025), by analyzing 10-K fillings, demonstrated that BertTopic model combined with FinTextSim embeddings outperforms significantly LDA and NMF topic models in identifying economic themes. However, as stated by the developers of the model (Grootendorst, 2022), the BertTopic model has the crucial limitation of being able to capture only one topic per input.

2.6 Defining strategic shifts

A strategic shift happens when a company fundamentally changes its direction or market focus to stay competitive. While traditionally viewed as one-off events, modern research shows that boards of directors actually see these changes as a series of connected moves (Klarner, Yoshikawa & Hitt, 2023). Accurately measuring and recognizing these shifts is important because they can have a direct impact on business potential and thus firm value. Historically, standard financial metrics have done a poor job of capturing the real value of these complex changes. For example, old financial models and research often suggested that diversification hurt a company's stock price ("the investors can diversify themselves"). Today, advanced machine learning and natural language processing tools help identify the moment changes take place, such as diversification. By using algorithms to quickly analyse many yearly reports, the company's true strategic direction can be observed. This new approach has proven that expanding and diversifying may, in fact, increase a firm's value (Choi, Chung & Kim, 2021).

2.7 Mechanisms of change

Firms execute strategic shifts through various mechanisms. The most straightforward may be mergers and acquisitions (M&A), where an external company is acquired in order to diversify or force a strategic shift. This allows companies to acquire new technologies, enter new markets or consolidate market power in order to acquire economies of scale. Another option would be internal restructuring, where operations are revisioned in order to improve efficiency. This would for example be the case if a company were to start using AI and cut many jobs.

Divestments are another way of changing the direction of a firm, where management spins off or sells underperforming or non-core parts of the business in order to refocus on the firm's primary objectives or increase profitability. Companies can also choose to develop new technologies themselves or be early adopters in other new technologies, such as the internet or AI.

These mechanisms are also commonly combined. This leads to an interrelated portfolio of options of strategic actions governed by the board of directors over time (Klarner et al., 2023). Succesful execution of a change in company direction fundamentally alters the

firm's risk profile and future cash flow capabilities. This leads to scrutiny by shareholders, which is why directors and management have to justify these actions and disclose them in certain reports.

2.8 Symbolic management and the hype phenomenon

When managers face pressure to change course or want to boost stock price quickly, they may be tempted to turn to symbolic management and submit to the hype rather than actually commit to a strategic shift. Executives use carefully chosen language in their public disclosures to frame their strategic shifts, a process that allows them to appease different stakeholders while "decoupling" their stated goals from their real, underlying operations (Fiss & Zajac, 2006). In essence, companies can create a smokescreen by changing how they talk about their business without actually changing the business itself.

This behaviour often leads to a "hype" phenomenon, where adopting trendy buzzwords, hypes or even changing the company name (think .com bubble) is used to artificially inflate market interest. Interestingly, investors even seem to reward these purely cosmetic or concept-driven changes with short-term stock price increases, proving that markets can be quite easily swayed by corporate framing and broader sentiment rather than substantive financial reality (Hui & Tan, 2022). By manipulating the narrative in their filings, managers can attempt to manage public perception and (temporarily) boost firm value without actually making meaningful strategic changes.

2.9 Measuring shifts (NLP)

To effectively capture the increasingly complex modern strategic pivot points, researchers are increasingly using sophisticated textual analysis techniques, instead of simply relying on static financial ratios. SEC 10-K filings, especially the business description (item 1) and the management discussion and analysis (item 7) offer a good description of the management's strategic narrative. By applying natural language processing to these texts, one can construct dynamic industry classifications, and more interestingly, the changes in this classification over time (Hoberg & Phillips, 2016). This NLP-approach maps the evolution of corporate strategy better than traditional, rigid industry codes. Furthermore, analyzing the tone of these public filings provides powerful predictive signals regarding future operations. For example, using machine learning algorithms to evaluate the forward-looking statements within the MD&A section uncovers nuanced tonal shifts that significantly predict future firm performance and liquidity (F. Li, 2010). Ultimately this transition to NLP enables financial researchers to quantify the subtle, qualitative language that executives use to frame their corporate strategy, and thus predict how investors actually process these disclosures.

2.10 Market pricing and the lazy prices effect

Despite the amount of information embedded in corporate filings, the stock market often fails to efficiently and accurately process textual changes. While investors may react quickly to surface-level hype and buzzwords, seemingly rewarding hollow strategic announcements with immediate, short-term stock price hikes (Hui & Tan, 2022), they seem to underreact to deeper substantive modifications deep inside official SEC filings. This behavioural anomaly is captured as the "lazy prices" phenomenon, which demonstrates that investors are generally not attentive to textual shifts within regular financial disclosures (Cohen et al., 2020). Specifically, when management alters the language in the risk factors (item 1A) or the MD&A (Item 7), these modifications serve as highly predictive signals of future changes in firm operations.

Because investors do not immediately price in these textual changes, firms that significantly alter their reporting language often experience predictable, long-term abnormal returns once the true underlying value in those changes reveal themselves (Cohen et al., 2020). Therefore, by tracking these specific reporting discrepancies, traders using machine learning models can attempt to exploit this market inefficiency and try to accurately forecast this long-term abnormal performance.

Overall, the past literature suggests that corporate disclosures contain multiple layers of information which are not fully reflected in market prices, and can be captured using specific approaches.

3 Data

Data and Summary Statistics

3.1 10-K Forms

For the collection of the data, our objective is to extract 10-Ks from companies in the time that they were part of the S&P 500. For this, a python script is deployed that extracts components of the 10-K that are needed for our study, rather than the entire 10-K. This is done to optimalize our storage and speed of using this data in our research. The parts extracted are Item 1, Item 1A and Item 7. The content of these items and their significance in our study are as follows:

Item 1 This section requires management to provide a detailed description of the company's operations, products, services, and the markets in which it competes. It includes information regarding the firm's business model, competitive positioning, and intellectual property. Item 1 serves as the primary Semantic Signal for the firm's Strategic Intent. Within the narrative of this section, management explicitly defines what the company

does and how that focus is evolving. By applying our all-MiniLM-L6-v2 architecture to this item, we can quantify Strategic Semantic Drift. For example: A transition in the thematic clusters of Item 1 from Hardware Manufacturing to Artificial Intelligence represents a measurable strategic pivot.

Item 1A In this section, companies are mandated to disclose the most significant factors that make an investment in the company speculative or risky. These disclosures range from macro-economic concerns and regulatory hurdles to specific operational risks. Item 1A represents the Linguistic Effect of a strategic shift. We use this section as our primary Dependent Variable for Risk. By applying the Loughran and McDonald (2011) financial dictionary to this item, we calculate Uncertainty Density and Litigious Sentiment. This allows us to observe whether a strategic pivot (as detected in Item 1) is followed by a Thematic Migration of risk. For example, we test if a shift toward high-tech strategies results in a statistically significant spike in technological or regulatory risk disclosures.

Item 7 This item provides a narrative explanation of the firm's financial performance from the perspective of management. It focuses specifically on liquidity, capital resources, and results of operations, often providing a bridge between the financial statements and the narrative strategy.

3.2 Financial Data

The financial data is extracted from two WRDS databases using the API function in Python. Accounting variables, including revenue, total assets, net income, R7D, EBIT and cash, are sourced from Compustat. Book equity is also extracted from Compustat. Stock market data, including returns, share price and shares outstanding, are sourced from CSRP in the period 2006 - 2025.

The two databases are then linked at the firm-year level using the 8-digit CUSIP identifier, which is common to both Compustat and CSRP. Market value is calculated as the product of share price and shares outstanding. Momentum is computed with the compounded return of months t-12 to t-2. We skipped the most recent month to avoid short-term reversal effects. We use the log form of the market value to comply with Fama-French standards.

3.3 Control variables

To ensure that the predictive power of the semantic drift score is not driven by known firm characteristics, we employ a proven set of control variables in our firm-level cross-sectional regressions. The foundational dataset relies on standard CUSIP firm identifiers and structural industry classifications to track firms and control for unobserved industry

heterogeneity. Raw accounting figures and performance metrics, including `revenue`, `cash` holdings, and the annual stock return (`annual_ret`) serve as the baseline data. By incorporating these variables, we aim to isolate the marginal effect of strategic textual shifts on subsequent stock returns from standard operational outcomes.

Building upon the Fama and French (1993) three-factor model and Carhart (1997)'s extension, we construct firm-level characteristics to control for size, value and momentum effects. Firm size is controlled for using the natural logarithm of market capitalization (`log_market_cap`), scaling the raw `market_cap` to account for extreme positive skewness in firm size. The value premium is traditionally captured using the book-to-market ratio, which is derived using the firm's market equity and underlying accounting figures. Additionally, to account for the tendency of winning stocks to continue performing well, we include a momentum control (`momentum`), calculated as the compounded firm return from month $t - 12$ to $t - 2$.

To align with the more extended Fama and French (2015) five-factor framework, we expand our control variables to account for profitability and investment patterns, which are significant drivers of cross-sectional expected returns. Profitability is captured using core operating metrics such as earnings before interest and taxes (`ebit`) and `net_income`, which can be scaled by `total_assets` to create a measure of operating profitability. To control for the investment factor, we monitor the growth in `total_assets`, while additionally incorporating research and development expenses (`rd_expenses`) to proxy for intangible capital investments. Controlling for these factors helps isolate the effect of strategic shifts, as firms undergoing these shifts might simultaneously exhibit temporary profitability shocks or heavy capital expenditure.

Comparing the baseline three-factor controls to the augmented five-factor specification provides a robustness check for our NLP signal. While the standard size, value, and momentum controls address the most prominent market anomalies, the addition of profitability and investment variables accounts for the underlying financial mechanics that often accompany strategic repositioning. By controlling for these fundamental financial metrics, we can conclude whether the semantic drift observed in 10-K filings contains unique, unpriced forward-looking information or if it is merely a textual reflection of existing accounting phenomena.

4 Methodology

4.1 Strategic Semantic Shifts via Sentence Embeddings

To detect year-over-year shifts in the deployed strategy of a firm, we apply the *all-MiniLM-L6-v2* sentence transformer model (source) to Item 1 of each 10-K filing. This model

Table 1: Descriptive Statistics

Variable	**Obs**	**Mean**	**Std. Dev.**	**p10**	**Median**	**p90**
Drift: Item 1	111023	0.0244	0.0229	0.0074	0.0176	0.0467
Drift: Item 1A	99825	0.0227	0.0252	0.0066	0.0149	0.0439
Drift: Item 7	111259	0.0271	0.0623	0.0054	0.0162	0.0357
Drift: Composite	112240	0.0262	0.0327	0.0091	0.0180	0.0416
Monthly Return	112035	0.0096	0.0844	−0.0909	0.0106	0.1071
Momentum (12–2m)	111992	0.1182	0.3001	−0.2393	0.1044	0.4750
Reversal (1m)	112188	0.0104	0.0784	−0.0849	0.0120	0.0999
log(MktCap)	112041	9.8156	1.1439	8.4274	9.7015	11.4319
Book-to-Market	108717	0.4647	0.3809	0.1018	0.3587	0.9534
ROA	111887	0.0653	0.0674	0.0056	0.0565	0.1514

Notes: This table reports summary statistics for the main variables used in the analysis. Drift variables are constructed from textual sections of 10-K filings. Returns are monthly.

Table 2: Descriptive Statistics: Fama-French Six-Factor Returns

Variable	**Obs**	**Mean**	**Std. Dev.**	**Min**	**Max**
Market Excess Return (MKT)	249	0.811	4.413	−4.886	5.602
Size Factor (SMB)	249	−0.007	2.664	−3.218	3.284
Value Factor (HML)	249	−0.079	3.163	−3.250	3.488
Profitability Factor (RMW)	249	0.357	1.866	−1.962	2.480
Investment Factor (CMA)	249	−0.014	1.885	−1.992	2.076
Momentum Factor (UMD)	249	0.108	4.370	−4.384	4.320
Risk-Free Rate (RF)	249	0.128	0.155	0.000	0.400

Notes: Monthly Fama-French six-factor returns in percent.

Table 3: Descriptive Statistics: Return Variables

Variable	**Obs**	**Mean**	**Std. Dev.**	**Min**	**Max**
Monthly Return	7583	0.014	0.078	−0.077	0.109
Monthly Return (%)	7583	1.416	7.765	−7.721	10.877
Momentum (12-2m)	2283	0.179	0.253	−0.094	0.496
Reversal (1m)	7462	0.014	0.077	−0.077	0.108
Annual Buy-and-Hold Return	1165	0.189	0.267	−0.103	0.519

Notes: Summary statistics for firm-level return variables.

Table 4: Sample Coverage by Year

Year	Firms	Filings	Mean Assets	Median Assets	Mean ROA
2004	462	462	33897	10077	0.0516
2005	462	463	36135	10294	0.0631
2006	465	465	37763	11151	0.0720
2007	475	484	41519	10671	0.0741
2008	491	491	41197	11320	0.0699
2009	493	492	34727	9790	0.0460
2010	485	485	38945	10746	0.0584
2011	484	483	39662	11409	0.0753
2012	485	484	40955	12877	0.0743
2013	487	487	44235	13534	0.0694
2014	486	485	44606	14809	0.0727
2015	490	490	48675	16370	0.0703
2016	499	498	47904	16099	0.0583
2017	498	498	51562	17812	0.0589
2018	501	501	51837	18705	0.0631
2019	500	499	51725	18743	0.0783
2020	496	495	57287	19738	0.0677
2021	500	500	59494	20278	0.0546
2022	499	498	62989	23040	0.0825
2023	497	496	63212	24889	0.0774
2024	500	499	66592	25683	0.0768

Notes: Total firm-filings: 10,278 — Total firm-months: 8,367 — Unique CIKs: 897.

generates dense 384-dimensional document embeddings that encode semantic meaning rather than surface-level word frequency. It is specifically trained and optimized for semantic textual similarity tasks using contrastive learning objectives, making it well-suited for comparing the strategic narrative of the same firm across consecutive years. Two filings that discuss the same strategic concepts, but with different vocabulary, will be mapped to nearby regions of the embedding space. Two filings that share the same vocabulary, but describe fundamentally different strategies will be mapped far apart. This model is therefore best suitable for our research.

Our choice of *all-MiniLM-L6-v2* over domain specific alternatives, like FinBERT, stems from the objective of the model. FinBERT is used for short-sentence financial sentiment classification, which is a different task from document-level semantic similarity. Applying a model like FinBERT to long-form business descriptions would risk combining tonal signals with strategic content and could potentially introduce noise. *all-MiniLM-L6-v2* is optimized to preserve semantic similarity structure, so that we can directly interpret the cosine distances as measures of strategic repositioning.

Since the MiniLM-L6-V2 model has a maximum input limit of 512 tokens and the section 1A and Section 7 filings typically exceed this limit, each document is split as described above into fixe size chunks, and for each of those chunk k a 384-dimensional embedding is generated.

The documents embedding is computed as the mean of all chunk embeddings as follows:

$$V_t = \frac{1}{K} \sum_{k=1}^{K} \text{embedding}(Chunk_k)$$

The semantic drift score is defined as:

$$\text{SemanticDrift}_{i,t} = 1 - \frac{\mathbf{v}_{i,t} \cdot \mathbf{v}_{i,t-1}}{\|\mathbf{v}_{i,t}\| \cdot \|\mathbf{v}_{i,t-1}\|} \tag{1}$$

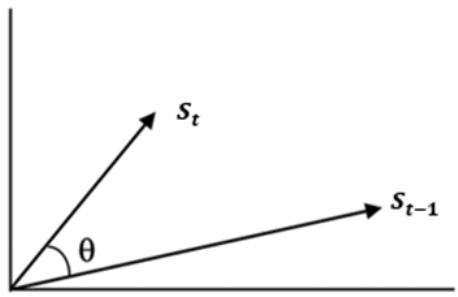

Figure 1: Angle between two 2-D vectors V_t and V_{t-1}

A value of 0 indicates that the firm's Item 1 embedding is identical across the two consecutive years, meaning that there is no strategic shift. A value of 1, or close to 1,

means that the two consecutive years are on different regions of the embedding space. This can be interpreted that there is indeed a strategic shift between the years. With the value of 1 indicating a complete strategic repositioning.

Compared with the alternative simple word-based similarity measures that has been used by Hoberg and Phillips (2016), our approach focuses on the meaning of text rather than simply matching words, making it more effective in capturing semantic divergence in the 10K-S disclosures.

The choice of cosine similarity over alternative distance metrics, such as Jaccard similarity or Euclidean distance, is motivated by its ability to focus on the orientation of embedding vectors within the semantic space, rather than their absolute magnitude Upadhyay and Khandelwal (2022). Specifically, while cosine similarity isolates the semantic consistency of a firm's disclosures across consecutive year—remaining invariant to differences in text length or the scale of numerical values.Euclidean distance is sensitive to the absolute differences in embedding coordinates. Furthermore, Jaccard similarity is considered a less suitable metric in this context, as it primarily captures word or token overlap rather than nuanced semantic meaning Upadhyay and Khandelwal (2022)

4.2 Portfolio Sorts

In order to evaluate whether the NLP strategic shift signal $SemanticDrift$ contains critical information about future stock returns, we employ a portfolio sorting approach following Cohen et al. (2020). At the end of each fiscal year, all S&P 500 firms with a valid signal observation are ranked and assigned to quintile portfolios Q_1 through Q_5. The Q_1 quintile contains the firms with the lowest signal values while Q_5 contains those with the highest. Each firm is then held in its assigned quintile portfolio for the twelve months following the fiscal year-end. Monthly portfolio returns are computed as both equal-weighted averages of the constituent stock returns, which are retrieved from CRSP-Compustat database.

Towards this direction, we regress using an extended Fama and French (1993) model following Carhart (1997) by adding a momentum factor (MOM) to the three-factor specification. Explaining the above, the market factor controls for aggregate market risk, SMB captures return differences related to size between small and large firms, and HML accounts for the value premium associated with high compared with low book-to-market firms. With the addition of momentum, we try to control for the possibility of past winners continue to outperform and past losers to underperform.

$$R_t^{LS} - R_t^f = \alpha + \beta_1 \cdot \text{MKT-RF}_t + \beta_2 \cdot \text{SMB}_t + \beta_3 \cdot \text{HML}_t + \beta_4 \cdot \text{MOM}_t + \varepsilon_t \qquad (2)$$

All factor data are obtained from Kenneth French's data library. The intercept α in each specification is the risk-adjusted abnormal return and represents the component of the long-short spread unexplained by any of the included systematic factors. Statistical inference is based on Newey–West heteroskedasticity and autocorrelation consistent standard errors. We avoid using a Fama-French six-factor model, as our objective is to avoid potential multicollinearity and overfitting derived from profitability and investment factors.

4.3 Cross-Sectional Regressions

The central question to be answered is whether firms that have experienced a strategic shift between the years exhibit differences in risk-adjusted returns in the year following their 10-K filing. While the portfolio sorts provide an intuitive test regarding the relation between $SemanticDrift$ and future returns, they do not control for specific firm characteristics known to predict the cross-section of returns. The same question is also applied to the broader set of text-change signals, where Cohen et al. (2020) predict that larger disclosure changes signal negative forward-looking information that is not yet reflected in prices.

In order to examine if $SemanticDrift$ predicts future company's returns in the cross section, we implement the Fama and MacBeth (1973a) regression procedure. For each month t, we estimate the cross-sectional regression of firms' excessive returns, based on the lagged signal:

$$R_{i,t} - R_t^f = \alpha_t + \beta_t \cdot \text{SemanticDrift}_{i,r(t-1)} + \delta_t' X_{i,t-1} + \varepsilon_{i,t} \tag{3}$$

where $R_{i,t}$ is the monthly return of firm i in month t, R_t^f is the risk-free rate, and $\text{SemanticDrift}_{i,r(t-1)}$ is the drift score measured from firm i's 10-K filing, as described above. The signal is derived as soon as a new 10-K is published, and is used to predict the returns in the following twelve months. The one-year lag is therefore a core assumption of the design, ensuring that that only information available to the public at $t-1$ is used to predict returns at time t. In that way, look-ahead bias is avoided, ensuring that no forward looking information enters the regression, avoiding look-ahead bias. The coefficients γ_t capture the cross-sectional pricing of firm-level factor exposures, ensuring consistency with the Fama-French factor structure. The vector $X_{i,t-1}$ includes the standard firm characteristics used and risk exposures. They are divided into three categories, the core return predictors(size, BtM, momentum), textual controls (sentiment of change,doc size, difference in doc length) and accounting controls(FCF, Accruals, Gross Profit, Investments). Finally, δ_t is the corresponding coefficient vector.

The coefficient of primary interest is β, which measures the marginal predictive power

of the signal for future excess returns after controlling for firm characteristics and systematic risk indicators in the cross-section. It captures whether firms with larger semantic shifts in their disclosures earn different subsequent returns. A negative and significant β, consistent with the finding of Cohen et al. (2020), would indicate that firms whose disclosures change more earn lower future returns.

In the second step, the estimated coefficients β_t are averaged over time to obtain the overall coefficient:

$$\bar{\beta} = \frac{1}{T} \sum_{t=1}^{T} \beta_t \tag{4}$$

This average coefficient indicates the expected cross-sectional relationship between the signal and future returns. Statistical inference is based on time-series standard errors with Newey and West (1987) adjustments to account for potential heteroskedasticity and autocorrelation.

A high value in SemanticDrift indicates that there is a difference in the way the company describes itself from the past year or its strategy, operations or outlook has shifted in a important grade. The above may reflect an entry into new markets, changes in business model, restructuring operations, or even a reaction to a shock. In short, it indicates a strategic instability or a repositioning. From an economic perspective, such changes may signal an increased uncertainty about their outcome. So, when a negative relation between semantic drift and subsequent returns are observed, indicates that investors might be skeptical about placing their bets on the company. The increased risk of this transition can make the returns lag, until positive results will be demonstrated. On the other hand, a positive relation would indicate that a strategic shifts created changes that increased company's value,and was recognized by the market at the upcoming period. Concluding, the regression tests whether such textual changes regarding the strategy shift are priced by the market.

5 Results - The results and tables are indicative only and may contain errors.

5.1 Portfolio Sorts

In the following section, the results of the portfolio sorting analysis designed to demonstrate the economic significance of textual signals for strategy shift are presented. From the reports of the four factors' (MKT, SMB, HML, UMD) alphas for equal-weighted and long-short portfolios (Table 1), we conclude that none of the abnormal returns created

from the above three signals are statistically significant. These portfolios were formed by sorting S&P 500 firms annually into quintiles based on the Semantic Drift Score.

Semantic Drift produces the largest raw alpha among the three signals, having a value of +0.00% per month (+0.00% annualised). Yet, the t-statistic of 0.61 is far below the conventional significance thresholds. The factor structure gives more evident compared to the other two signals. the portfolio loads negatively on the market factor ($\beta_{\text{MKT}} = -0.00$) and positively on size and value ($\beta_{\text{SMB}} = +0.00$, $\beta_{\text{HML}} = +0.00$), generating an R^2 of 0.097. The negative market beta indicates that the long-short portfolio is short on high-beta stocks. However, after controlling for these systematic exposures the intercept remains insignificant.

Table 5: Carhart Four-Factor Alphas for Long-Short Portfolios

Signal	$\hat{\alpha}_m$	$\hat{\alpha}_a$	t-stat	MKT	SMB	HML	UMD	R^2	N
Semantic Drift	+0.00%	+0.00%	(0.00)	−0.00	+0.00	+0.00	−0.00	0.000	000

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. - No signal has statistically significance.
Firms are sorted annually into quintiles on each signal. Long-short portfolios are equal-weighted Q5 minus Q1. Monthly excess returns are regressed on the Carhart (1997) four-factor model. $\hat{\alpha}_m$ and $\hat{\alpha}_a$ denote monthly and annualised alphas respectively. Newey-West HAC standard errors with 6 lags.

5.2 Fama-MacBeth Cross-Sectional Regressions

Table 6: Fama-MacBeth Cross-Sectional Regressions

	(1) Signal only		(2) + Firm chars		(3) + Risk exp	
Variable	Coef.	t-stat	Coef.	t-stat	Coef.	t-stat
Signal = Semantic Drift	*T = 0 / 0 / 0 months, Avg N = 0 / 0 / 0*					
Intercept	0.000*	(1.00)	0.000	(0.00)	0.000**	(0.00)
Semantic Drift (std)	0.000	(0.00)	0.000	(0.00)	−0.000	(−0.00)
Firm Size (log)			−0.000*	(−0.00)	−0.000*	(−0.00)
ROA			0.000	(0.00)	0.000	(0.00)
Leverage			0.000	(0.00)	−0.000	(−0.00)
Book-to-Market			−0.000	(−0.00)	−0.000	(−0.00)
Momentum (12-2m)					−0.000*	(−0.00)
Reversal (1m)					0.000	(0.00)
Avg R^2	0.000		0.000		0.000	

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.
Each month a cross-sectional OLS regression is estimated. Time-series averages of monthly coefficients ×100 are reported (monthly return in percent). Newey-West t-statistics with 6 lags in parentheses. Signals are cross-sectionally standardised each month.

To extend our evidence, we examine whether the textual signals are priced in the cross-section of stock returns. For that reason, we use the Fama and MacBeth (1973b)

regression framework, as described earlier. In Table 2, we can find the report of the Fama-MacBeth cross-sectional regression results for all three signals, across three progressively richer specifications. In the first one, we have the signal alone, next, the signal with firm characteristics (size, ROA, leverage, book-to-market), and finally the full specification adding risk exposures (momentum and reversal). Coefficients represent monthly return in percent. Newey-West t-statistics, computed with 6 lags and adjusting the standard errors, are reported in parentheses.

Semantic Drift. In the Panel are presented the results occurred for the $SemanticDrift$, which is the main signal of interest. Its coefficient is positive specifications (1) and (2), having values of $+0.0786$ ($t = 0.39$) and $+0.1712$ ($t = 0.48$) respectively, but turns slightly negative when risk exposures are added (-0.1068, $t = -0.32$). Yet, None of the above estimates is statistically significant.

Regarding the controls for $SemanticDrift$, firm size is negative and significant in specifications (2) and (3) ($\beta = -0.3317$, $t = -2.10$ and $\beta = -0.4275$, $t = -2.09$). This indicates the presence of a size premium within the sample. Momentum is also significant and negative in specification (3) ($\beta = -1.2888$, $t = -2.04$). This result has the opposite sign compared to the other two panels.

6 RENS RESULTS

6.1 Portfolio sorts

6.2 Fama–MacBeth Regressions (Per Signal)

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Table 7: Return Predictability and Factor Loadings

	Item 1	**Item 1A**	**Item 7**	**Composite**
Panel A: Long-Short Alphas Across Factor Models				
CAPM	-0.1791*	-0.3590***	-0.1978	-0.2041**
	[-1.94]	[-2.70]	[-1.60]	[-2.14]
FF3	-0.1932**	-0.3275**	-0.2458***	-0.2179**
	[-2.31]	[-2.43]	[-2.58]	[-2.56]
FF3 + Mom	-0.1737**	-0.3179**	-0.2286**	-0.1990**
	[-2.02]	[-2.45]	[-2.41]	[-2.35]
FF5	-0.2250***	-0.3682***	-0.2825***	-0.2525***
	[-2.69]	[-2.92]	[-2.91]	[-3.03]
N	239	230	239	239
Panel B: Factor Loadings (FF3 + Momentum)				
α	-0.1737**	-0.3179**	-0.2286**	-0.1990**
	[-2.02]	[-2.45]	[-2.41]	[-2.35]
MKT	0.030	-0.034	0.067	0.036
	(1.34)	(-1.78)	(2.61)	(1.48)
SMB	0.012	0.013	0.012	0.028
	(0.37)	(0.32)	(0.30)	(0.72)
HML	-0.121	0.112	-0.293	-0.134
	(-3.49)	(2.44)	(-7.61)	(-3.37)
UMD	-0.064	-0.029	-0.056	-0.062
	(-3.29)	(-1.18)	(-2.07)	(-2.29)
R^2	0.101	0.056	0.297	0.111
N	239	230	239	239
Panel C: Factor Loadings (FF5)				
α	-0.2250***	-0.3682***	-0.2825***	-0.2525***
	[-2.69]	[-2.92]	[-2.91]	[-3.03]
MKT	0.056	-0.005	0.087	0.065
	(2.95)	(-0.28)	(3.55)	(2.63)
SMB	0.046	0.044	0.050	0.062
	(1.34)	(0.99)	(1.39)	(1.71)
HML	-0.126	0.032	-0.284	-0.161
	(-3.34)	(0.65)	(-5.30)	(-2.80)
RMW	0.068	0.033	0.099	0.055
	(1.29)	(0.51)	(1.86)	(0.95)
CMA	0.055	0.219	0.009	0.112
	(0.93)	(2.95)	(0.12)	(1.57)
R^2	0.080	0.086	0.290	0.101
N	239	230	239	239

This table reports long-short portfolio alphas and factor loadings for drift-based strategies. Panel A reports alphas across asset pricing models. Panels B and C report full factor loadings. Alphas are monthly returns multiplied by 100. Factor loadings are in levels. Newey–West (6 lags) t-statistics are reported in brackets or parentheses.
***, **, * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Table 8: Quintile Portfolio Sorts

Signal	**Portfolio**	**Mean**	**Ann.**	**Vol.**	**t**
Item 1	Q1	0.890	11.215	15.792	3.02
	Q5	0.875	11.021	16.392	2.86
	Q5–Q1	−0.015	−0.176	4.710	−0.17
Item 1A	Q1	1.072	13.644	16.619	3.39
	Q5	0.806	10.109	16.579	2.60
	Q5–Q1	−0.247	−2.918	6.124	−2.11
Item 7	Q1	0.916	11.560	15.595	3.14
	Q5	0.895	11.284	15.959	3.00
	Q5–Q1	−0.021	−0.250	5.787	−0.19
Composite	Q1	0.939	11.866	15.564	3.23
	Q5	0.904	11.409	16.183	2.99
	Q5–Q1	−0.034	−0.412	4.865	−0.38

Notes: Portfolios are formed monthly based on quintiles of drift measures. Returns are equal-weighted. Values are in percentage terms.

Table 9: Fama–MacBeth Regressions (Per Signal)

	Item 1		Item 1A		Item 7		Composite	
Variable	**U**	**C**	**U**	**C**	**U**	**C**	**U**	**C**
Item 1	−0.0068	-0.0366*						
	(-0.31)	(-1.89)						
Item 1A			-0.0482**	-0.0451*				
			(-2.00)	(-1.74)				
Item 7					−0.0142	−0.0231		
					(-0.60)	(-1.18)		
Composite							−0.0232	−0.0328
							(-1.04)	(-1.64)
Momentum		-0.5512*		-0.6683**		-0.5333*		-0.5350*
		(-1.91)		(-2.22)		(-1.87)		(-1.87)
Reversal		−0.9383		-1.2594*		−0.9395		−0.9331
		(-1.43)		(-1.72)		(-1.44)		(-1.41)
log(MktCap)		0.3159***		0.3370***		0.3154***		0.3165***
		(6.60)		(6.55)		(6.59)		(6.62)
Book/Market		-2.2480***		-2.2496***		-2.2110***		-2.2309***
		(-10.48)		(-9.71)		(-10.11)		(-10.34)
ROA		-5.5138***		-5.4047***		-5.4853***		-5.5142***
		(-6.99)		(-6.41)		(-6.86)		(-6.90)
Intercept	0.9542***	−0.7898	0.9322***	−1.0156	0.9556***	−0.8023	0.9551***	−0.8019
	(3.31)	(-1.25)	(2.92)	(-1.50)	(3.32)	(-1.27)	(3.31)	(-1.27)
Months (T)	239	239	215	215	239	239	239	239
Avg N	463	449	462	447	464	450	468	454
Avg R^2	0.0029	0.0918	0.0032	0.0949	0.0028	0.0907	0.0027	0.0912

Notes: Coefficients are multiplied by 100. t-statistics (Newey–West with 6 lags) are in parentheses. ***, **, * denote significance at the 1%, 5%, and 10% levels, respectively.

6.3 return predictability of different holding periods

Table 10: Return Predictability by Holding-Period Horizon

Signal	Horizon	LS Spread	t	FM β	t	α (Carhart)	t
Item 1	1m	0.5832	1.12	−0.0457	−0.29	0.8091*	1.77
	3m	0.0157	0.07	−0.1195	−1.22	−0.0064	−0.03
	6m	0.1783	1.02	−0.0326	−0.59	0.1044	0.51
	12m	−0.0155	−0.17	-0.0342*	−1.73	-0.1838**	−2.14
Item 1A	1m	−0.3313	−0.72	−0.0041	−0.02	−0.2558	−0.61
	3m	−0.2273	−0.77	0.0228	0.19	−0.2440	−0.88
	6m	−0.1545	−0.79	-0.1232*	−1.85	−0.2110	−0.95
	12m	−0.1880	−1.64	-0.0513*	−1.94	-0.2516*	−1.97
Item 7	1m	−0.0127	−0.02	−0.5379	−0.99	−0.2495	−0.48
	3m	0.2276	0.80	−0.1981	−0.85	−0.0026	−0.01
	6m	0.1335	0.87	−0.1283	−1.11	−0.1124	−0.80
	12m	−0.0443	−0.41	−0.0266	−1.39	-0.2523***	−2.66
Composite	1m	0.1933	0.39	−0.1886	−0.75	0.1722	0.35
	3m	−0.1895	−0.69	−0.0935	−0.72	−0.3176	−1.00
	6m	−0.0718	−0.45	−0.1011	−1.01	−0.2372	−1.29
	12m	−0.0669	−0.74	-0.0378*	−1.91	-0.2339***	−2.80

Notes: This table reports return predictability across different holding-period horizons. LS Spread denotes the return difference between the highest and lowest drift quintiles. FM β is the Fama–MacBeth coefficient standardized by one standard deviation of the signal. α is the Carhart four-factor alpha. All return measures are monthly percentages. ***, **, * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

6.4 Interaction effects

Table 11: Fama–MacBeth Regressions with Interaction Terms

Signal	**Interaction**	β**(Drift)**	**t**	β**(Interact)**	**t**	**T**	**N**
Item 1	Size	-0.0454**	−2.14	0.0593**	2.32	239	449
	Book/Market	-0.0415**	−2.08	-0.0889***	−3.32	239	449
	ROA	-0.0873**	−2.56	0.7280*	1.89	239	449
Item 1A	Size	-0.0447*	−1.69	0.0044	0.16	215	447
	Book/Market	−0.0414	−1.51	-0.0534**	−2.06	215	447
	ROA	−0.0653	−1.51	0.4427	0.75	215	447
Item 7	Size	−0.0229	−0.67	0.0659*	1.86	239	450
	Book/Market	−0.0443	−1.39	-0.1634***	−3.04	239	450
	ROA	−0.0338	−0.83	0.1547	0.26	239	450
Composite	Size	−0.0362	−1.53	0.0311	1.22	239	454
	Book/Market	−0.0222	−1.01	-0.1075***	−4.10	239	454
	ROA	−0.0569	−1.63	0.4985	1.04	239	454

Notes: This table reports Fama–MacBeth regressions with interaction terms between drift signals and firm characteristics. Coefficients are multiplied by 100. t-statistics (Newey–West with 6 lags) are reported in parentheses. ***, **, * denote statistical significance at the 1%, 5%, and 10% levels, respectively. A negative interaction with size indicates stronger effects among smaller firms, while a positive interaction with book-to-market indicates stronger effects among value firms.

7 Conclusion

The results across both the portfolio sorts and the Fama-MacBeth regressions are consistent. The $SemanticDrift$ signal does not generate statistically significant return predictability in our sample of S&P 500 firms over the 2005 - 2024 period.

The lack of significance in the drift signal comes in contrast with Cohen L. (2020), which in the paper show a negative relation between disclosure changes and upcoming returns. One possible explanation is that our sample is restricted to S&P 500 firms. These companies are heavily covered by analysts and institutional investors so any information regarding the changes in the annual disclosures is likely to be quickly incorporated into prices. Therefore, there is not much space left to identify any late price reactions promptly. This explanation is also consistent with the efficient markets hypothesis fama1970efficient, as applied to large-cap stocks.

Future work should take the analysis further and ckeck also for firms beyond the S&P 500. There, gaps in information are more likely to allow textual signals indicate better any delayed returns.

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